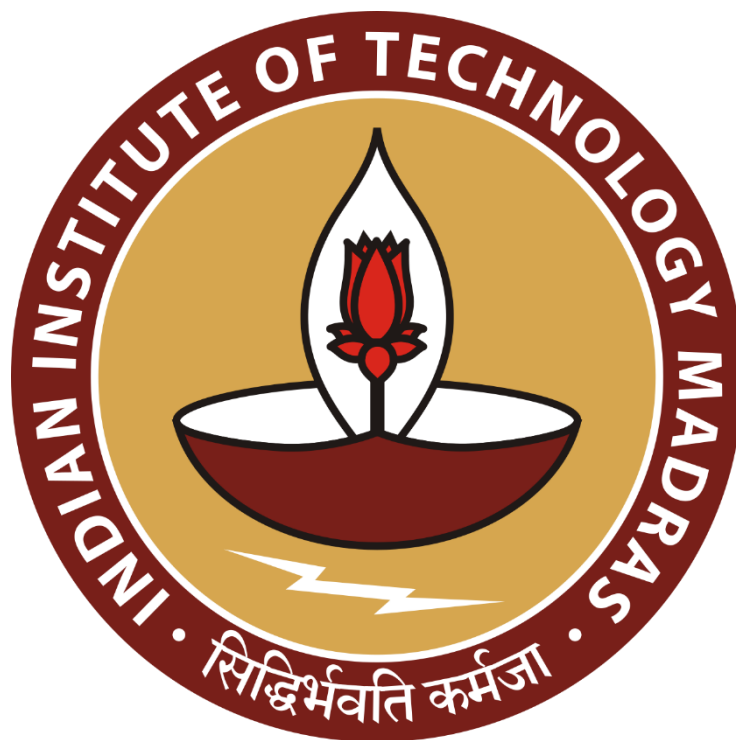




Milestone 2 Report: Enter the Transformers



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Milestone 2 Report – Enter the Transformers

Objective

The objective of Milestone 2 was to transition from traditional Natural Language Processing techniques to Transformer-based language models. This milestone introduced the Hugging Face ecosystem and demonstrated how pre-trained Transformer models can generate context-aware representations for text without requiring task-specific training. The work also explored the architecture of BERT, the concept of attention mechanisms, sentence embeddings, and zero-shot classification for multiple-choice question answering.

Work Completed

1. Environment Setup

The required Transformer libraries were installed and configured, including:

- Hugging Face Transformers
- Hugging Face Datasets
- Sentence Transformers
- Accelerate

These libraries enabled loading of pre-trained language models and simplified working with transformer-based NLP pipelines.

2. Context-Aware Sentence Embeddings

A pre-trained Sentence Transformer model (all-MiniLM-L6-v2) was loaded to generate dense semantic embeddings.

Tasks completed:

- Generated embeddings for sample sentences.
- Produced contextual embeddings for prompts and all answer options in the training dataset.
- Stored embeddings for later similarity computation.

Unlike TF-IDF and Word2Vec used in Milestone 1, these embeddings capture the contextual meaning of words and sentences.

3. Semantic Similarity using BERT Embeddings

Cosine similarity was computed between the prompt embedding and each answer option embedding.

The implementation:

- Calculated similarity scores for all answer choices.
- Ranked answer options according to semantic similarity.
- Generated Top-3 predictions for every question.
- Evaluated prediction quality using the mAP@3 metric.

This demonstrated how Transformer embeddings can be directly applied to retrieval-style multiple-choice problems.

4. BERT Tokenization

The notebook introduced the BERT tokenizer by:

- Loading the bert-base-uncased tokenizer.
- Tokenizing sample questions.
- Examining token IDs.
- Converting token IDs back into readable tokens.

This helped explain how raw text is converted into model-readable input.

5. BERT Model Architecture

A pre-trained BERT model was loaded to understand its internal working.

The notebook demonstrated:

- Encoding text into hidden representations.
- Extracting the final hidden states.
- Using the [CLS] token representation as the sentence embedding.
- Understanding embedding dimensions produced by the model.

This section introduced the practical workflow of obtaining contextual sentence representations directly from BERT.

6. Understanding the Attention Mechanism

The milestone introduced the Transformer attention mechanism conceptually.

Topics covered include:

- Self-attention.
- Context-aware word representations.
- How each token attends to every other token.
- Why Transformers outperform traditional sequential models for language understanding.

This formed the theoretical foundation behind BERT and RoBERTa models.

7. Zero-Shot Classification

The Hugging Face Zero-Shot Classification pipeline was explored.

Experiments included:

- Classifying prompts into general academic subjects.
- Testing improved candidate labels.
- Using answer options themselves as candidate labels.

- Producing Top-3 predictions without any additional model training.

The zero-shot predictions were also evaluated using the previously implemented mAP@3 metric.

8. Softmax vs Sigmoid

The notebook compared two commonly used activation functions.

The implementation demonstrated:

- Softmax for mutually exclusive class prediction.
- Sigmoid for multi-label classification problems.
- Practical PyTorch examples illustrating the differences between both approaches.

9. Prompting Small Language Models (SLMs)

The notebook introduced prompting techniques for Small Language Models.

This section demonstrated how instruction prompts can be used to obtain predictions directly from a language model instead of relying only on embedding similarity or supervised training.

10. Milestone 2 Assignment Questions

The notebook concludes with implementations for the milestone assessment.

Solutions include:

- Loading datasets using the Hugging Face Datasets library.
- Tokenizing text with BERT.
- Extracting contextual embeddings.
- Computing semantic similarity.
- Performing zero-shot classification.
- Generating model outputs using Transformer-based sequence-to-sequence models.
- Producing answers required for the milestone evaluation.

Outcomes

By completing Milestone 2, the project successfully transitioned from classical NLP techniques to modern Transformer-based Natural Language Processing. Context-aware embeddings significantly improved semantic understanding compared to traditional vector representations. The notebook demonstrated how pre-trained Transformer models can perform feature extraction, semantic matching, and zero-shot classification with minimal additional training.

The milestone also established the foundation required for subsequent stages involving fine-tuning Transformer models such as BERT, RoBERTa, and DeBERTa for improved multiple-choice question answering performance.